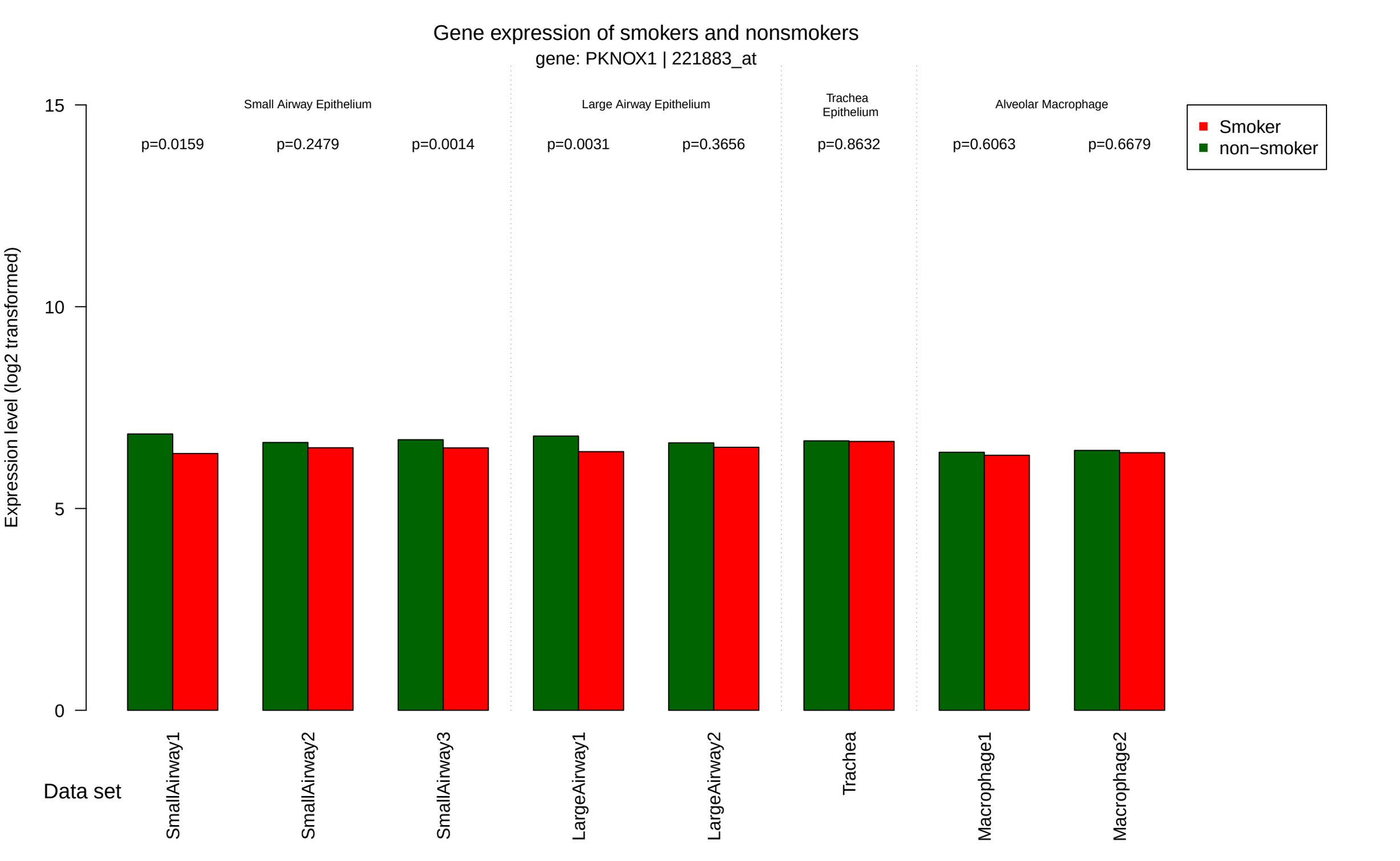
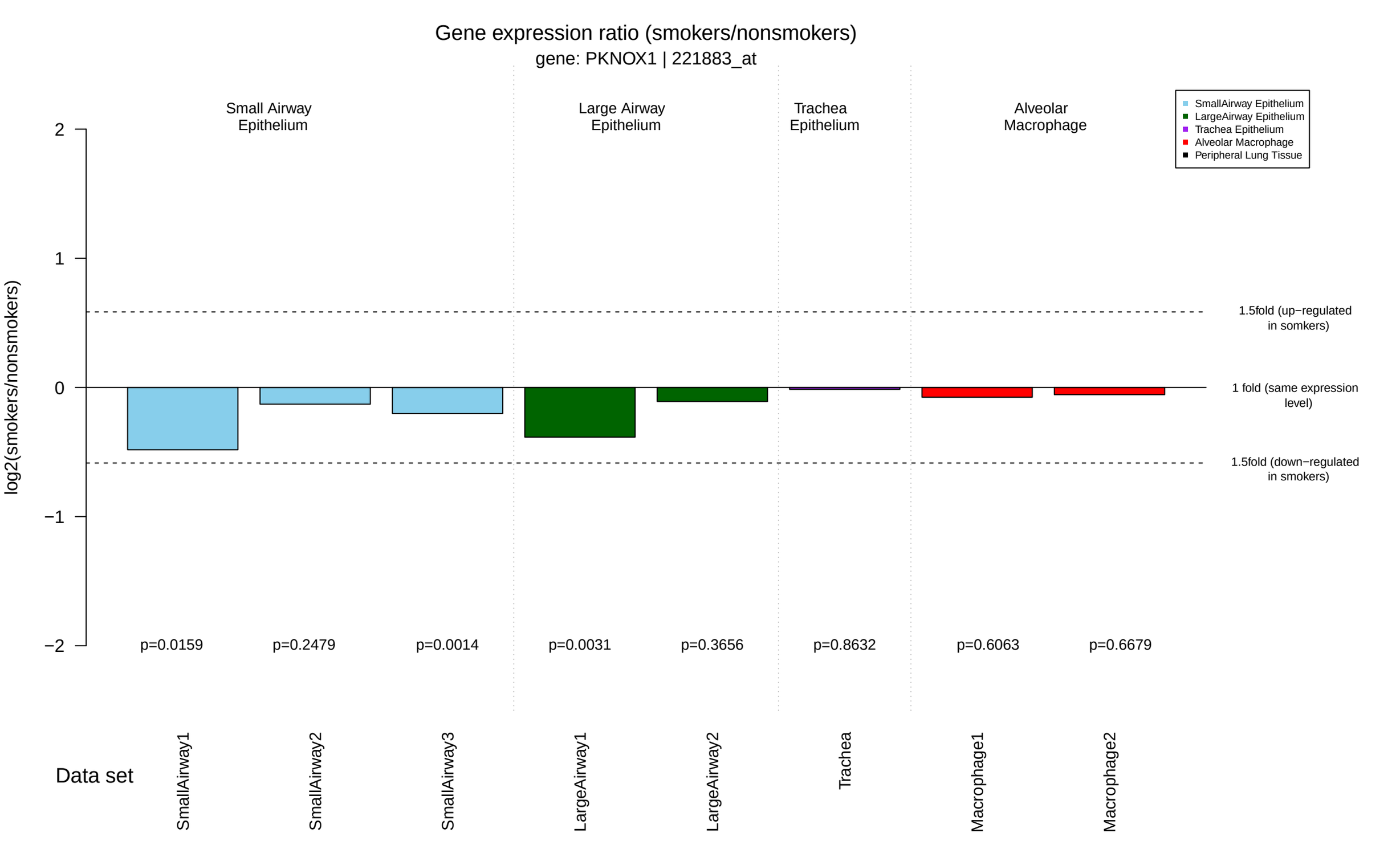
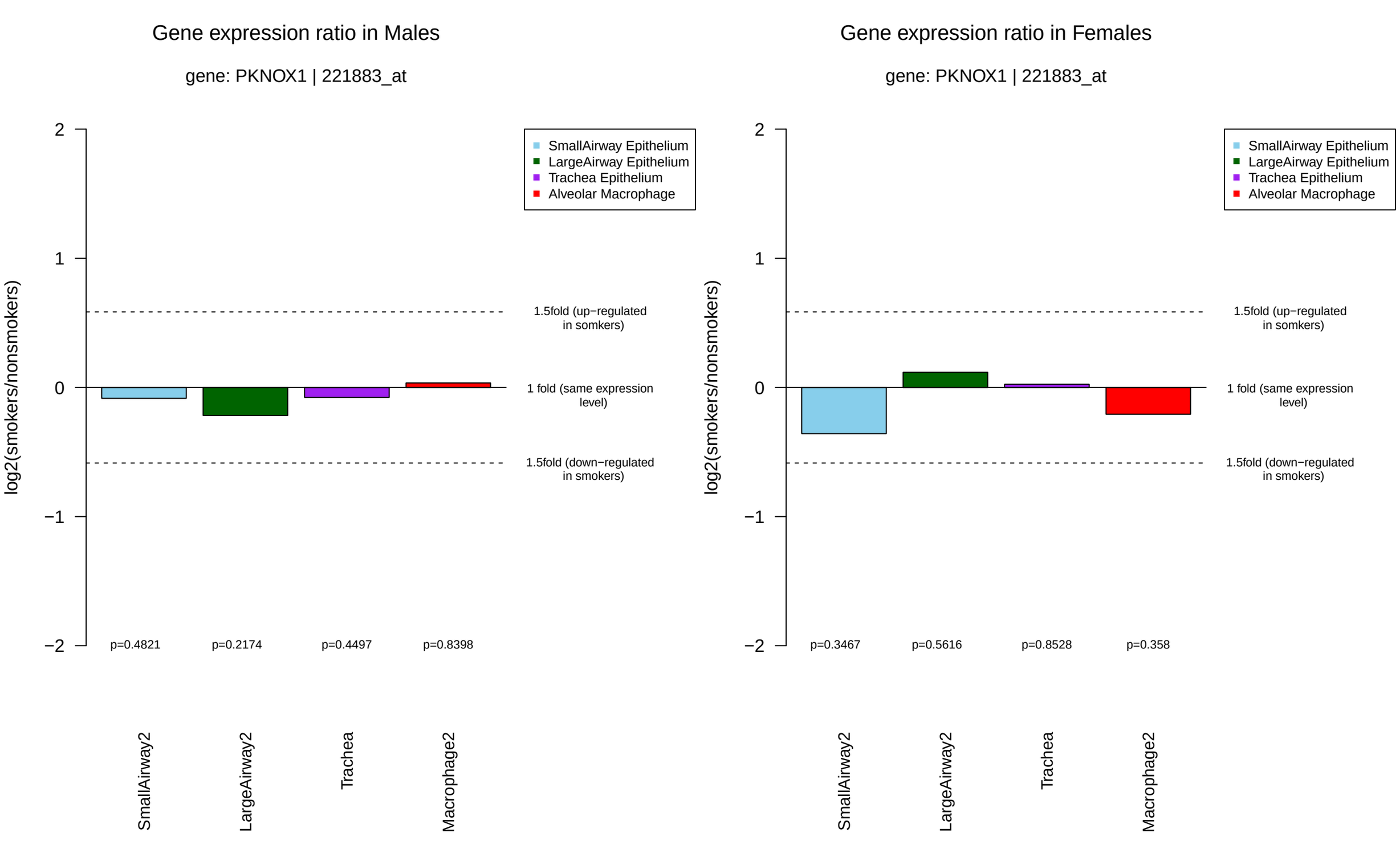
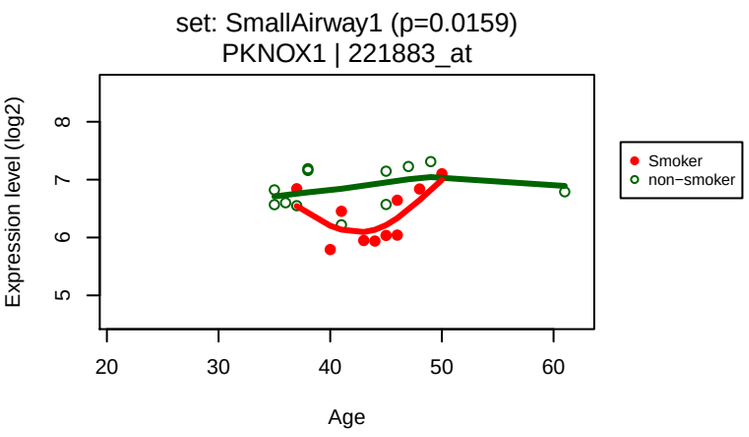
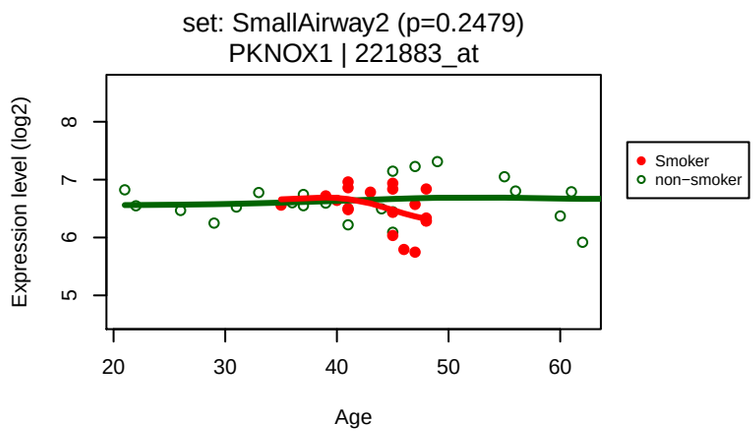
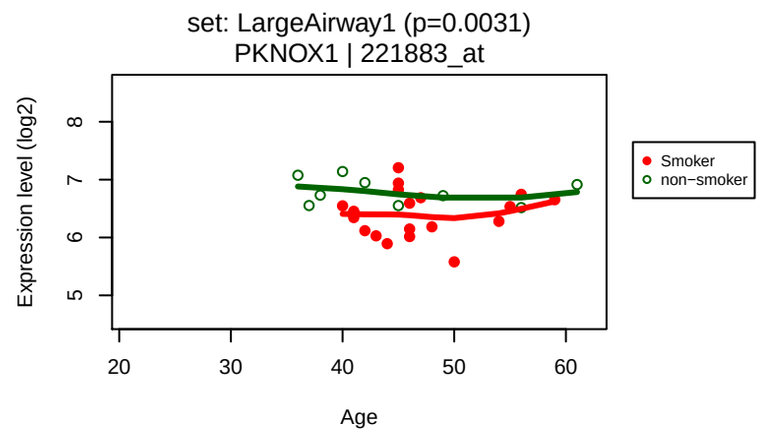

Gene expression of smokers and nonsmokers
gene: PKNOX1 | 221883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0159
p=0.2479
p=0.0014
p=0.0031
p=0.3656
p=0.8632
p=0.6063
p=0.6679
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

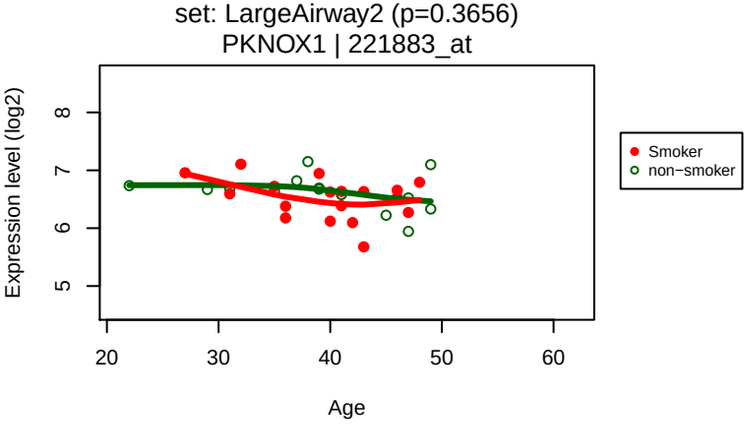

Gene expression ratio (smokers/nonsmokers)
gene: PKNOX1 | 221883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0159
p=0.2479
p=0.0014
p=0.0031
p=0.3656
p=0.8632
p=0.6063
p=0.6679
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

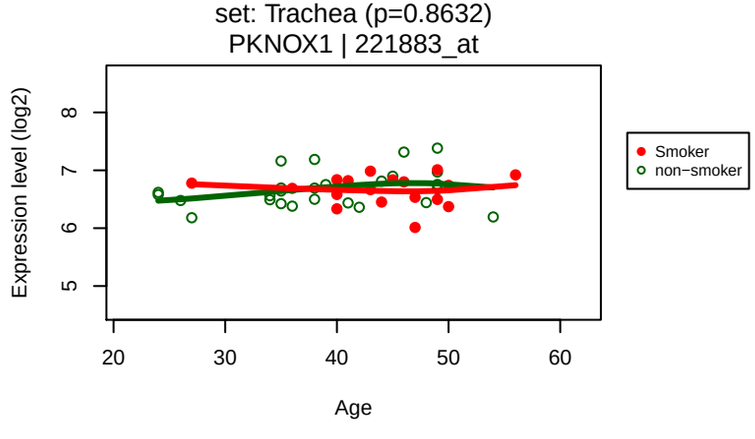

Gene expression ratio in Males
gene: PKNOX1 | 221883_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4821
p=0.2174
p=0.4497
p=0.8398
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PKNOX1 | 221883_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3467
p=0.5616
p=0.8528
p=0.358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

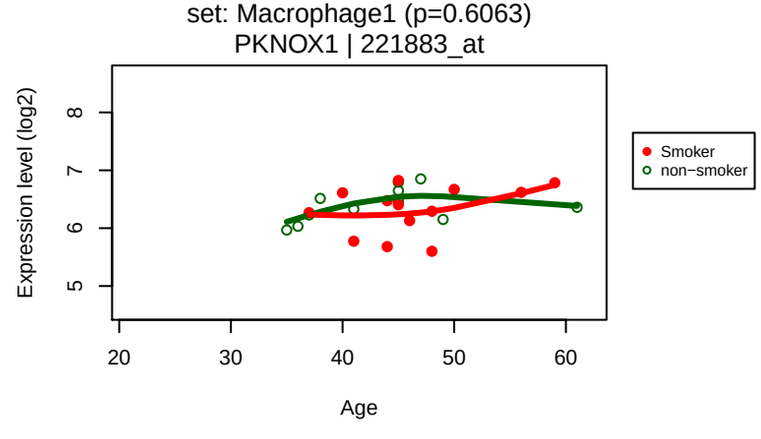

set: SmallAirway1 (p=0.0159)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker

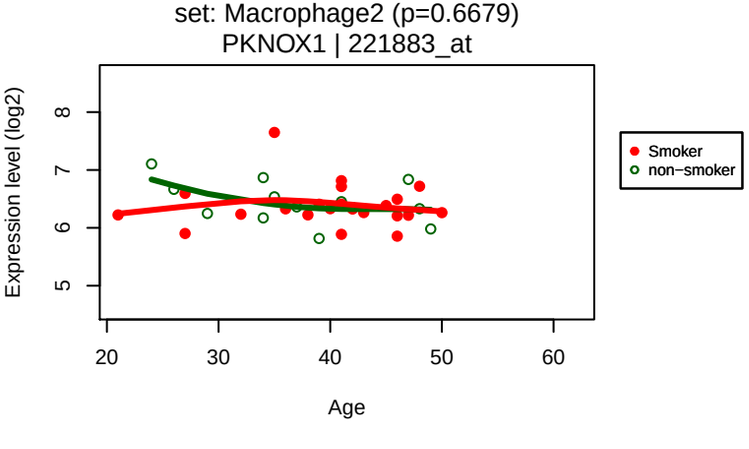

set: SmallAirway2 (p=0.2479)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker

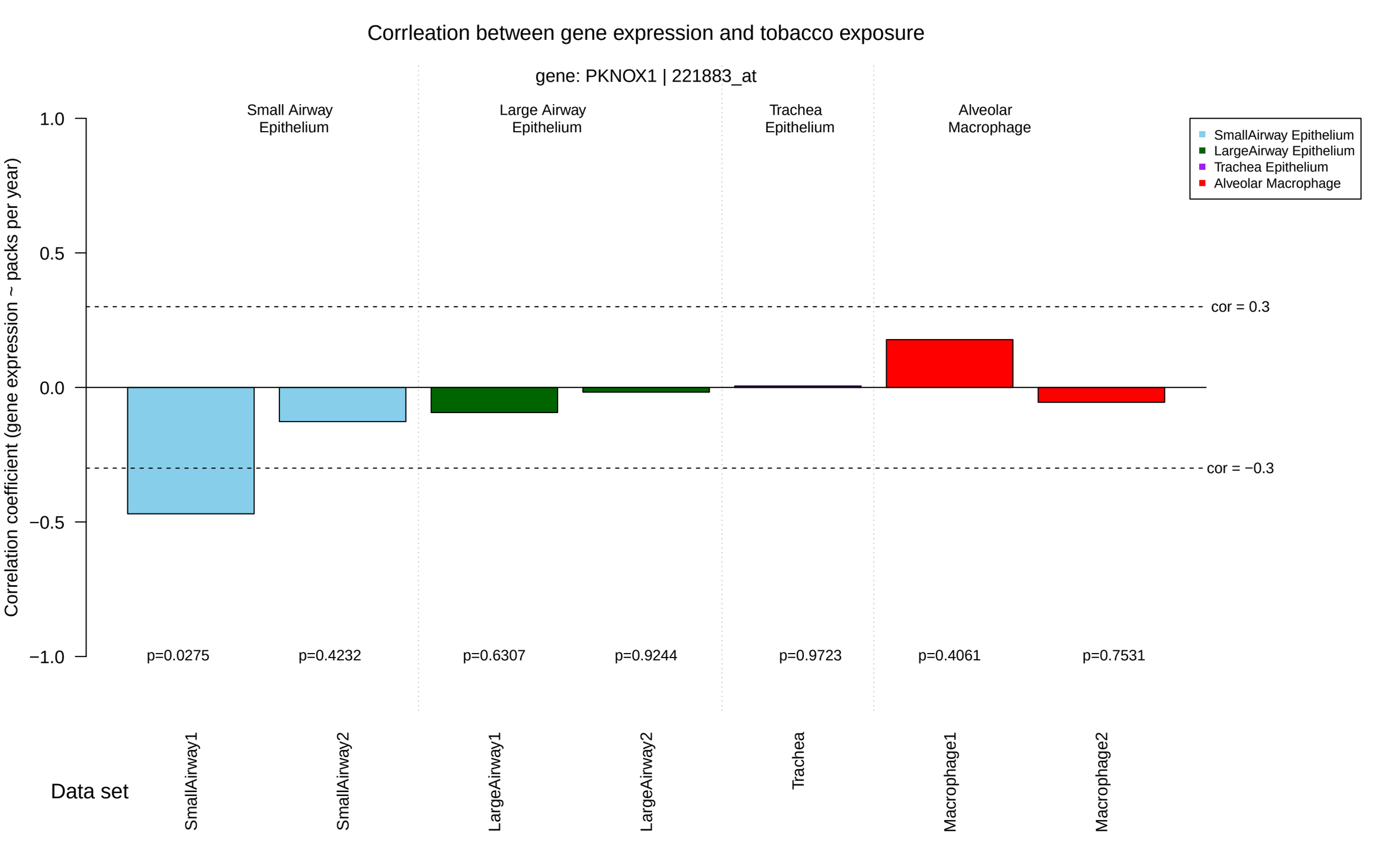

set: LargeAirway1 (p=0.0031)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3656)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8632)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6063)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6679)
PKNOX1 | 221883_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PKNOX1 | 221883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0275
p=0.4232
p=0.6307
p=0.9244
p=0.9723
p=0.4061
p=0.7531
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2